



**School of Journalism
and Mass Communication**
UNIVERSITY OF WISCONSIN-MADISON

J677: Concepts and Tools for Data Analysis and Visualization Section 004

Fall 2026 (3 credits)

Tuesday: 11:00am-12:15pm

Thursday: 11:00am-12:15pm

Meeting Room: Vilas 5055

Instructor: Yoo Ji, Suh

Office: Vilas 5007

Email: yooji.suh@wisc.edu

Office Hours: Tuesday 2:00-3:00pm or by appointment

Course Designation and Attributes: Advanced Undergraduate/Graduate Lecture and Lab

Instructional Mode: In-person

How Credit Hours Are Met by the Course

Students are expected to attend the scheduled class meeting each week, and work on course learning activities (reading, writing, problem sets, group project, etc.) outside class for about 6 hours in total. About half of the out-of-class hours should be devoted to hands-on exercises and group projects.

Course Description

Like at no other time, our world is recorded in digital format. The information generated by this ceaseless recording is the raw material for a new era of learning about and analyzing everything: from storm systems to diseases to individuals' interactions with friends and relatives and even their physical movements. The result is that researchers, journalists, strategic communicators, companies and organizations are now inundated with data, and rapidly innovating new ways to create, access, and analyze those data. There is great need for people who are able to use digital information to investigate important questions, draw useful insights, and communicate those insights to others—very often in compelling visual format—while also being realistic and honest about what data can and cannot tell us.

That is what this class is for: it is an introduction to the world of data and how data can be used to answer questions that interest journalists, strategic communicators, and students of social sciences in general. This course emphasizes the critical connections between data analyses that would involve statistical processing of raw data and visualization strategies that would facilitate faithful, accurate, and compelling delivery of data insights to targeted audiences. It offers a combination of conceptual training, instruction in specific tools for data analysis and visualization, and the opportunity to put new skills to use in a final project.

This class is designed for students at the upper undergraduate levels in journalism, strategic communication, and other social science majors who have interest in using data to advance their academic and/or professional goals. **Master's students, especially those in professional degree programs**, may also find the course useful (please note that the course is not an advanced research

methods course as our orientation to data is much more descriptive and focused on storytelling than hypothesis-testing and/or causal inference). This course emphasizes applications of data analyses and visualizations in practical settings. For students who absolutely do not want to engage with data or coding, this course might not be a good fit. If you feel unsure, please feel free to discuss it with me.

This is a hybrid concepts and skills course. Readings, lectures, discussions and assignments will deal with **concepts** related to data analysis and **software skills** involving data manipulation, analyses and visualization. The aim is to foster the ability not just to churn out visualizations, but to **think** meaningfully, scientifically and creatively about the stories that lie within data.

The structure of the course follows its hybrid nature: we will typically use the first half of the week **for either lecture or seminar discussion, and the second half for lab time, hands-on exercises and assignments**. We will thus often be working on two parallel tracks simultaneously: one conceptual and one skills-based.

Course Learning Outcomes

Students completing the course will:

- Be aware of practical and ethical challenges of collecting, managing, analyzing and presenting data
- Understand the basic grammar and principles for data visualization
- Be familiar with the R language for managing data and creating visualizations
- Be comfortable with basic statistical concepts in data analysis and visualization
- Develop a working knowledge of handling and visualizing tabular and textual (e.g., social media posts) data for evidence-based storytelling

Technical Abilities

We will focus mainly on **R (and RStudio)**. We will be using *Posit Cloud*, a cloud-based browser interface to R coding, to handle assignments but you are encouraged to install R and RStudio on your own computer. Both are open-source and free.

This course is tailored to students with backgrounds in the social sciences (not engineering, not computer science). No familiarity with programming or advanced statistics is expected. The most important thing for you to bring to the course is **curiosity** and **persistence**: the willingness to think about interesting questions and how they might be answered with data, and the patience to endure trial and error in answering them, including a learning curve necessary to pick up programming skills in R.

To be sure, there will be math and statistics and coding, which are indispensable to data analyses and critical to visual communicators in today's digital age. We will discuss and use basic concepts in statistics and related reasoning skills needed to treat data appropriately and identify misuses of data. But our approach is fundamentally hands-on: it is not about abstract equations or solving numeric puzzles but about why uses of numbers and data are appropriate in certain instances and not in others. This is intended to give the course an intuitive foundation.

Finally, you will be learning some new software in the course, as you have done many times before. The goal here is twofold: one is simply that there are certain tools that are very useful and good to

have in your personal toolkit. At the same time, tools do change quite rapidly--so the second goal is for you to develop the skill of picking up new software skills fairly rapidly. Students who already have experience with design, image manipulation or web publishing software will find ways to apply those skills in the course.

Graduate Student Notes

It is likely that graduate students will have specific interests or goals to pursue, and class content can be modified to suit those goals: for example, a student may wish to focus on learning Python in place of R. Some graduate students will already be familiar with statistical concepts well above the level we discuss. **If you already are comfortable with any of the concepts or tools we are using, please talk to me in advance of those assignments and deadlines so that we create a plan that is productive for you.**

When it comes to the final project, graduate students are expected to create a product of especially high quality that is related to a scientific research study. Graduate students' final project requirements will include a longer paper, more analyses, deeper discussion of key concepts and visualization choices, and integration with the student's larger course of study.

All graduate students should meet with the instructor at least once before Week 8 to discuss how the final project fits with their program.

Study Strategies

This is a concept-skill hybrid class and the end goal is to help you appreciate the power of data visualizations and leverage modern data processing tools such as R towards a project of your choice. Coding is a necessary component of this process. A few tips to help you succeed if you are new to coding.

- **Learn by doing.** Nobody can pick up coding just by reading books or tutorials. Practice, practice, and practice. We have many in-class coding exercises as well as assignments to help you establish the routine to practice coding on a regular basis.
- **Practice regularly.** Like learning dancing or swimming, you will need a lot of frequent practicing to overcome the initial steep learning curve. I will strongly discourage you from waiting until the last minute to complete assignments. Coding is not something you could easily pick up or make up by merely staying up overnight. Designating a regular block of time (e.g., one hour every other day) to go through lab notes, follow textbook tutorials, and complete assignments will be a way more efficient strategy.
- **Take advantage of your study groups.** You will be part of a study group throughout the semester to help each other and complete group assignments. Group members are a great resource for debugging your codes, catching up with new R commands, and asking questions. Of course, when it is your turn, please be a giver!

Working with AI Agents

This course adopts a "co-piloting" mindset and approach, encouraging smart and responsible use of AI agents. I do not ban AI use categorically, but I also do not think blindly outsourcing assignments to AI would help learning. The real question is how to effectively and responsibly incorporate AI

into your learning and workflows. The principles below explain my reasoning, and the practical recommendations follow from them.

Principles and Rationale:

- **Learning needs "frictions"**. One downside of the complete outsourcing model is that AI will always produce an output, and that process is often too smooth to be conducive to learning. We learn from trial and error. It is worth noting that AI learns by reinforcement learning, which is itself a process that involves seeing examples of both successes and failures. If you route around every difficulty, you also remove the very experiences that learning depends on.
- **Humans ultimately sign off on AI outputs**. Humans need to be held accountable for research outputs, regardless of AI involvement. When we write a data analysis report, regardless of how much AI helped with writing the analytical code or the text itself, we ultimately take responsibility for any claims and errors. The same logic applies to your assignments and your final project in this course.
- **Accountability requires the ability to audit**. Given the accountability requirement, we need to be able to audit AI outputs, and that would not happen unless we have learned to assess the quality of AI-generated code ourselves. Learning to assess code, in turn, requires having written a good amount of it on your own. This is the practical reason why the "frictions" above are worth preserving.
- **AI benefits are multipliers, not substitutes**. AI is most helpful to those who already have a solid basis. Its benefits multiply for seasoned users, but they could turn into penalties for newcomers who wish to forgo the "trial and error" process of learning. The gap between what you can generate and what you can evaluate is exactly where the risk lies, and outsourcing widens that gap.
- **Working effectively with AI is itself a skill, and it too rests on domain understanding**. How to design prompts, how to structure a workflow with AI (e.g., how to manage conversations to save token usage, how to create and use reusable skill files, how to set up a multi-agent system) are increasingly valuable skills. All of them, however, still require a solid understanding of the problem itself. You need to know enough to decompose a task, specify it clearly, and recognize when the output is wrong.

Practical Recommendations:

- **Always attempt to complete each assignment by yourself first**. When you hit a barrier, consult others, consult the internet, and of course, consult AI. The sequence matters more than the sources. You will also find that articulating a problem clearly enough to ask about it often resolves the problem itself.

- **Decide before you prompt.** Work out what estimand you want and what model corresponds to it, and then use AI to implement your decision.
- **Ask AI to explain, not just to produce.** Asking what specific codes do and why one package is chosen over another; asking for the code alone does not. Used well, AI can serve as a patient tutor that answers the same question five different ways at midnight.
- **Do not submit code you cannot explain.** Before any AI-assisted code goes into your assignment, make sure you can explain what every line does and why it is there.
- **Be skeptical when AI agrees with you.** Current AI systems tend to be agreeable and may validate a flawed analytical plan rather than push back.
- **Draft your own prose.** Writing is a form of thinking, and hence learning as well. You often do not know what you actually think about a problem until you try to write it down, and a sentence that refuses to come out right is usually a signal that the underlying idea is not yet clear. Outsourcing the draft means outsourcing that process. The interpretation sections of your assignments are where you develop and demonstrate your own understanding, so the words should be yours. AI is allowed and encouraged for copyediting and grammar correction, including punctuation, awkward phrasing, tightening a paragraph that runs long, and checking formatting. The line is between polishing prose you have written and generating prose you have not.
- **Disclose your AI use.** Include a brief note at the end of each assignment describing how you used AI tools, if at all. This is not punitive and will not affect your grade. It helps me see where these tools are helping and where they may be substituting for understanding. In your AI disclosure statement, specify:
 - Tool & Version: (e.g., Opus 4.8, ChatGPT 5.6).
 - Description: How you used the tool (e.g., code checking and explanation, editing, summarizing, formatting).
 - Example: “This syllabus was developed with AI assistance (Claude Opus 5), used for checking and formatting citations, editing an initial draft from my own notes and prior syllabi, and thinking through assignment decisions. All substantive choices, including the reading list and course structure, are mine, and I have verified every claim and citation.”

Finally, note that the final project includes an **oral exam component**, where we will ask you to explain the R codes submitted for the final project. The exam serves as a check on whether the understanding has actually transferred, and if you have worked with AI in the ways recommended here, you should find it a fair one.

Required Textbooks

- Healy, K. (2026). *Data visualization: A practical introduction (2nd ed.)*. Princeton University Press. <https://socviz.co/>
- Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science (2nd ed.)*. O'Reilly Media. <https://r4ds.hadley.nz/data-visualize.html>

Additional readings will be available on Canvas.

Grading

	<u>Points</u>
<u>Individual Assignments</u>	
# Lab 1: Intro to R	2
# Lab 2: Intro to Tidyverse	2
# Lab 3: Intro to ggplot2	2
# Lab 4: Bar Plots	2
# Lab 5: Data Sources	2
# Lab 6: Data Cleaning & Tidy Data	2
# Lab 7: Bivariate Scatter Plots	2
# Lab 8: Visualizing Uncertainty & Regression Models	2
# Lab 9: Choice of Label, Annotations, Scales, and Legends	2
# Lab 10: Choice of Color, Themes, and Visual Focus	2
# Lab 11: Text Mining	2
# Lab 12: Resume and Career Portfolio Integration	2
Subtotal:	24%
<u>Group Assignments</u>	
#0 Group Sign-Up and Contract	0
#1 The Best and the Worst of Data Visualization	10
#2 Tabular Data Visualization (Mid-Term Exam)	10
Subtotal:	20%
<u>Final Project</u>	
#0 Idea Pitch	1
#1 (Individual): Exploratory Analysis	10
#2 Cleaned Dataset	5
#3 Instagram Post	5
#4 Infographic	5
#5 AI Role Playing	5
#6 Executive Brief	5
#7 Final Poster	10
#8 Oral Exam	10
Subtotal:	56%

Bonus: Winners of the Top 3 posters

2

Total:

100%

Grading Scale

93 – 100 = A

88 – 92 = AB

83 – 87 = B

78 – 82 = BC

70 – 77 = C

60 – 69 = D

0 – 59 = F

Course Schedule*

Date	Topic	Readings	Assignments Due
WEEK 1 : Sep 3			
Sep 3	Overview of the course		
WEEK 2: Sep 8 & Sep 10			
Sep 8 Lab #1	Intro to R	Healy (2026): Chapter 2 A quick intro to R objects: https://rforcats.net/	
Sep 10 Lecture	Human vision & Key principles of visualization	Cairo (2013): Chapters 5-6 (On Canvas)	
WEEK 3: Sep 15 & Sep 17			
Sep 15 Lecture	How visualizations lie & other bad examples	Healy (2026): Chapter 1	Lab #1 Due Group Assignment #0: Sign-Up and Group Contract
Sep 17 Lab #2	Intro to Tidyverse		
WEEK 4: Sep 22 & Sep 24			
Sep 22 Lecture	The grammar of visualization I	Healy (2026): Chapter 3	
Sep 24 Lab #3	Intro to ggplot2		Lab #2 Due Group Assignment #1: The Best and the Worst of Data Visualization
WEEK 5: Sep 29 & Oct 1			
Sep 29 Lecture	The grammar of visualization II	Healy (2026): Chapter 4	
Oct 1 Lab #4	Bar Plots		Lab #3 Due
WEEK 6: Oct 6 & Oct 8			
Oct 6 Lecture	Data	Lazer & Radford (2017)	
Oct 8 Lab #5	Data Sources		Lab #4 Due
WEEK 7: Oct 13 & Oct 15			
Oct 13 Lecture	Data Wrangling		Final Project #0: Idea Pitch
Oct 15 Lab #6	Data Cleaning & Tidy Data		Lab #5 Due
WEEK 8: Oct 20 & Oct 22			

Oct 20 Lecture	Tabular data: Descriptive vs. Explanatory Stories	Healy (2026): Chapter 5	<i>Graduate students: please make sure you've discussed with me your final project ideas</i>
Oct 22 Lab #7	Bivariate Scatter Plots		Lab #6 Due
WEEK 9: Oct 27 & Oct 29			
Oct 27 Lecture	Tabular data: When Things Get Complicated	Cairo (2016): Chapters 4, 11 (on Canvas); Healy (2026): Chapter 6	
Oct 29 Lab #8	Visualizing Uncertainty & Regression Models		Lab #7 Due
Week 10: Nov 3 & Nov 5			
Nov 3 Lab #9	Choice of Label, Annotations, Scales, and Legends		Final Project #1 (Individual): Exploratory Analysis
Nov 5 Lab #10	Choice of Color, Themes, and Visual Focus	Healy (2026): Chapter 8	Lab #8 Due
WEEK 11: Nov 10 & Nov 12			
Nov 10 Lecture	Mid-Term Oral Practice & Feedback I		Group Assignment #2 (Mid-Term Exam): Tabular Data Visualization
Nov 12	Mid-Term Oral Practice & Feedback II		Lab #9 Due
WEEK 12: Nov 17 & Nov 19			
Nov 17 Lecture	Text mining	Text Mining with R: Chapter 1; Text Mining with R: Chapter 2; Andrews (2025a); Andrews (2025b)	Final Project #2: Cleaned Dataset
Nov 19 Lab #11	Text Mining		Lab #10 Due
WEEK 13: Nov 24 & Nov 26			
Nov 24 Lab #12	Resume and Career Portfolio Integration		Final Project #3: Instagram Post Lab #11 Due
Nov 26	No Class (Thanksgiving)		

WEEK 14: Dec 1 & Dec 3			
Dec 1	Final Project Check-In		Lab #12 Due
Dec 3			Final project #4: Infographic
			Final project #5: AI Role Playing
WEEK 15: Dec 8 & Dec 10			
Dec 8	Oral Exam & Poster Show		Final project #6: Executive Brief;
Dec 10			Final project #7: Final Poster;
			Final project #8: Oral Exam

*This represents a tentative schedule for the current semester. The instructor reserves the right to make changes as necessary.

References

- Andrews, M. (2025a, January 28). *Qualitative text analysis with local LLMs: Part I*.
https://www.mjandrews.org/notes/text_analysis_with_llms/part1.html
- Andrews, M. (2025b, January 28). *Qualitative text analysis with local LLMs: Part II*.
https://www.mjandrews.org/notes/text_analysis_with_llms/part2.html
- Cairo, A. (2013). *The functional art: An introduction to information graphics and visualization*. New Riders.
- Cairo, A. (2016). *The truthful art: Data, charts, and maps for communication*. New Riders.
- Healy, K. (2026). *Data visualization: A practical introduction (2nd ed.)*. Princeton University Press.
<https://socviz.co/>
- Lazer, D., & Radford, J. (2017). *Data ex machina: Introduction to big data*. *Annual Review of Sociology*, 43(1), 19–39. <https://doi.org/10.1146/annurev-soc-060116-053457>
- Silge, J., & Robinson, D. (2017). *Text mining with R: A tidy approach*. O'Reilly Media.
<https://www.tidytextmining.com/>
- Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science (2nd ed.)*. O'Reilly Media. <https://r4ds.hadley.nz/data-visualize.html>

Course Policies

Readings

The assigned readings for each class period must be read before that class. Come to class prepared – it is the only way to get the full benefit from the lectures.

Assignments

Each assignment is due on the designated date. Always keep a copy of your assignments for your records, and remember that it is your responsibility to see that the instructor receives your assignments. On an assignment due date, please come to class prepared to discuss the material covered in the assignment. If you struggled with an assignment, come prepared to ask questions.

Attendance and Participation

Regular attendance is important and mandated in this course. You will be developing an original visualization project throughout the course of the semester, which you simply cannot do if you miss class on a regular basis. A large portion of your grade will be based on group assignments, which require regular interactions among group members. I do not take attendance but typically there is a strong positive correlation between regular attendance and the final grade.

Usage of Audio Recorded Lectures Statement

Lecture materials and recordings for J677 are protected intellectual property at UW-Madison. Students in this course may use the materials and recordings for their personal use related to participation in this class. Students may also take notes solely for their personal use. If a lecture is not already recorded, you are not authorized to record my lectures without my permission unless you are considered by the university to be a qualified student with a disability requiring accommodation. [Regent Policy Document 4-1] Students may not copy or have lecture materials and recordings outside of class, including posting on internet sites or selling to commercial entities. Students are also prohibited from providing or selling their personal notes to anyone else or being paid for taking notes by any person or commercial firm without the instructor's express written permission. Unauthorized use of these copyrighted lecture materials and recordings constitutes copyright infringement and may be addressed under the university's policies, UWS Chapters 14 and 17, governing student academic and non-academic misconduct.

Email

Email submission of assignments will NOT be accepted (and will not be graded). Feel free, however, to email me with any questions, comments, or concerns about the class. Please note that I do not sit by the computer around-the-clock. That said, I will try to reply to most emails within 48 hours.

Students with Disabilities

Students with a McBurney VISA who need special accommodations for exams should notify the instructor by the end of Week 2 so that we can begin making accommodations. Do NOT wait until the last minute to make exam accommodations – it takes advance planning. If you need any other accommodations, please let the instructor know as soon as possible.

Academic Dishonesty

Unless otherwise stated, all work in this course is to be completed on an individual basis and

should be the original work of the student. This class will follow University and L&S guidelines concerning scholastic misconduct. If academic misconduct occurs, the minimum penalty for the student will be a zero on the assignment and a report filed at the Office of Student Conduct and Community Standards. Additional penalties may occur based on the seriousness of the offense.

It is the responsibility of each student to be aware of what constitutes academic misconduct. Examples include plagiarism, patchwork writing, cheating on exams, quoting sources word-for-word without the use of quotation marks, and other activities described in Wisconsin State Legislature UWS 14.03: https://docs.legis.wisconsin.gov/code/admin_code/uws/14/03.

As part of a broad range of educational tools used by SJMC to help ensure academic integrity, the course instructor is using Turnitin software. All indicated assignments from all students must be submitted to turnitin.com as part of the submission process for work completed in this class. The Turnitin.com tool then compares the work from one student with a variety of sources, including (but not limited to) websites, publications, and a database of prior papers. Turnitin then provides the instructor with links to possible matches between the student's paper and these other sources, along with a 'similarity score.' The instructor (not the software) then makes a complete assessment whether, or not, any form of academic misconduct may have occurred.

Creating an Inclusive Learning Environment in this Class

The issues we will discuss in this class cannot be thoughtfully and rigorously discussed without an occasional reference to unpopular ideas or to offensive material. Students and instructors alike are expected to remain sensitive to individual differences. The diversity of a multicultural society requires that we discuss differences with no anger, arrogance, or personal attacks, and without perpetuating stereotypes about gender, age, race, religious affiliation, sexual preference, national origin, dialect, or disability. This also goes for political affiliations. In this class, we must be aware of our own biases and privilege, avoid asking others to stand in for an entire community (even implicitly) or to have those people bear the burden of educating us. Please try to curtail your inevitable feelings of defensiveness. Recognize your feelings of angst etc. when they arise, interrogate them, and sit with them before acting on them as much as possible. Understand that no matter how we identify, we all occupy privilege (even just being a student at UW affords us a certain privilege that others do not have). We are all in this together, but let's respect where we all are on our individual journeys. By "safe" space, I mean a place where people feel comfortable expressing themselves, asking questions, and otherwise engaging in the class. However, "safe" does not mean banning different perspectives that you do not agree with. Statements of hate or intolerance – no matter who they come from – are not welcome in this class.

Policy on Generative AI Systems

You may use generative AI tools (e.g., Claude, ChatGPT, Gemini, Copilot, Kimi, DeepSeek, etc.) in this class unless otherwise specified, as long as you do so transparently and in ways that enhance rather than outsource your learning. The "Working with AI Agents" section above describes what this means for the assignments in this seminar. Specifics vary based on the learning goal. If you are not sure whether a given use is appropriate, please ask.

Transparency requirements. Clearly state AI use or non-use on all assignments. If you used AI, include a brief description of what you used (e.g., Claude Fable, GPT 5) and how (e.g., for debugging ggplot2 codes, for grammar check). Reflection on whether and how it was actually helpful is always welcome, and often more instructive than the assignment itself.

Plagiarism. AI use that is not disclosed is cheating. Plagiarism from AI, like plagiarism from any other source, is a violation of academic integrity principles, and it carries the same consequences: plagiarized content earns a zero and potentially more serious academic action. Let's avoid that. One concrete example: if you did not disclose using AI for references and your submission contains a hallucinated citation, the fabricated reference can be treated as conclusive evidence of undisclosed AI use, which is a far more serious matter than simply losing points for inaccurate citations.

How I will handle a suspected case. If I have reason to think AI use went undisclosed, I will ask you to meet with me and walk me through your submission — what the code does, why you made particular choices, where specific claims came from. Much like the final oral exam. If the work is yours, the conversation is short and that is the end of it. If it is not, that conversation establishes how much of the work was not your own, and I will proceed from there, up to and including a formal report to the Office of Student Conduct and Community Standards under UWS 14. I use a conversation rather than an AI detector because detectors produce false positives and cannot tell disclosed use from undisclosed use. A conversation can.

Accountability. You are responsible for all of your work, including any inaccurate, offensive, or otherwise unethical content, regardless of its source. This is the same accountability principle described above: authorship means signing off, and signing off means the errors are yours.

A closing thought. Efficiency in a course is often in opposition to genuine learning. What saves you time this semester may cost you exactly the understanding this seminar exists to build, and that trade is almost never worth it at this stage of your training. We are all still developing norms around AI in research and teaching, so let us do so in ways that cultivate trust between us and make our time in this seminar meaningful.